

## The five-factor gambling motivation model

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### Abstract

Pathological gambling is a serious social issue, but the underlying psychological motivations are poorly understood. This study developed a five-factor gambling motivation model. The five factors that motivate gambling (socialization, amusement, avoidance, excitement, and monetary motives) were derived from study data obtained from 240 college students. The structure of the five-factor model was confirmed by factor analysis of responses from 234 frequent gamblers. We then compared the “monetary motive version” of this model with the “parallel version” to determine which model more accurately describes how the five factors influence gambling. The monetary motive model holds that amusement, excitement, and avoidance motives influence gambling severity only through mediation of the monetary motive. The parallel model proposes that the five motives all independently influence gambling severity. We found that the avoidance and excitement motives did not have direct effects on gambling severity, but the monetary motive showed a direct positive influence. Thus the monetary motive model was more effective than the parallel model in explaining the influence of specific gambling motives on gambling severity. These findings may help in the improvement of therapy for pathological gambling.

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### 1. Introduction

Pathological gambling has become a serious social issue in Korea recently. About 3% to 4% of Koreans are estimated to be pathological gamblers (Lee et al., 1999; Kim et al., 2002). The fraction of pathological gamblers is relatively higher in Korea compared to the 1% to 2% of pathological gamblers in Canada (Ladouceur, 1996), the

United States (Shaffer et al., 1997; Volberg, 1996), and European countries (Beconia, 1996). Not only does gambling cause financial distress, it also predisposes to job loss, family conflict, illegal acts, depression, suicide and substance abuse, among other problems (Black and Moyer, 1998; Lesieur, 1992a; Lorenz and Yaffee, 1986; Rosenthal and Lorenz, 1992; Petry and Kiluk, 2002; Specker et al., 1996). For these reasons, it is important to explore why people are motivated by and involved in gambling.

In general, motivation is defined as the internal and/or external force that triggers, directs, intensifies, and leads to the persistence of a behavior (Weiner, 1980). Cooper et al. (1992) proposed a three-factor motivational model to

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explain why people drink alcohol. This model includes three motives: the coping (or avoidance) motive to avoid or manipulate negative emotions, the social motive for socializing and having fun with people, and the enhancement (or amusement) motive to intensify positive moods such as fun and enjoyment. This three-factor model for the motives underlying drinking behavior is mentioned in this article because this model can be applied to a gambling motivational model as well.

Recently, Neighbors et al. (2002) classified 16 gambling motivations and used these to develop the South Oaks Gambling Screen (SOGS). In a test of 184 college students, 79 (45.6%) participants were non-problem gamblers (i.e., SOGS=0), 72 (41.6%) had a minimal tendency toward gambling problems (i.e., SOGS of 1 or 2), 16 (9.3%) were subclinical gamblers (i.e., SOGS of 3 or 4), and 6 (3.5%) were probable gamblers (i.e., SOGS of  $\geq 5$ ); 11 participants did not complete the survey. The strongest motivations for gambling among the subjects were winning money (42.7%), having fun (23.0%), socializing (11.2%), and experiencing excitement (7.3%). In total, 84% of the participants pointed to winning money, enjoyment, socialization, or excitement as gambling motivations. In a random telephone survey of 1522 adults, the most frequently reported reason for gambling was winning big money (50.5%). The next most commonly reported reasons were entertainment (33.4%), excitement (18.4%), curiosity (10.6%), and socialization (9.5%) (Mississippi State University Gambling Group, 1995). Klingemann (1995) suggested that gambling is a form of entertainment conducted for fun and enjoyment. Altogether, these findings suggest that entertainment and socialization motives, which also play important roles in alcoholism, are very important gambling motives for non-problem gamblers. Furthermore, they suggest that excitement and financial gain are powerful and unique motives in gambling.

These studies were limited, however, by the fact that the research was mostly performed on social gamblers. In 1996, Coman et al. (1997) conducted a study about the reasons for gambling by polling callers to G-line. G-line is a confidential telephone service that offers counseling and treatment referrals for problem gamblers. The aggregate results revealed that 42.0% of males reported gambling for financial reasons, 29.5% reported excitement, 22.6% did it for pleasure, and 18.4% engaged in it for stress reduction. In the case of women, they gambled due to boredom (36.5%), loneliness (31.3%), and stress reduction (24.6%). They also gambled for pleasure (22.6%) and financial reasons (21.7%). Jacobs (1987) reported similar results. When asked to rank, in order, a range of possibilities for their gambling activities,

pathological gamblers reported stimulation, relaxation, and escaping from emotional tension as motives for their gambling. The next most frequent gambling motivations they reported were fleeing from a painful reality, escaping from an unhappy home, escaping from an unhappy job, and escaping from a humdrum life. When asked to indicate the three most important influences that led to their continued gambling, they stated thrill of the action, wanting to make a lot of money quickly, and simply not being able to stop. Other research indicated that the motives for habitual gambling were winning money and spending time (8.5%), socialization (33%), and feeling thrill and excitement (50%) (Commission on the Review of the National Policy Toward Gambling, 1976). Problem gamblers engaged in gambling to avoid negative feelings, boredom, and low self-esteem (Blaszczynski and Steel, 1998; Raghunathan and Pham, 1999). These results suggest that the avoidance motive (especially for women), like in the three-factor motivational model of alcoholism, exerts an important role in the continuation of gambling behavior, and the excitement and monetary motives are very important in problem gambling.

Neighbors et al. (2002) stated that identifying motives for gambling among individuals who fall across the entire continuum is necessary to better understand what differentiates problem from non-problem gamblers, and that motivational differences are primarily important in degree versus type of gambling. Based on previous empirical research about gambling motivations, gambling motives across the entire continuum can be summed up as five motives: socialization, amusement, excitement, avoidance, and monetary. The avoidance, amusement and socialization motives have already been validated for alcoholism (Cooper et al., 1992).

Recent studies on gambling motives have focused on the direct relationship between gambling severity and gambling motives, but they have neglected to mention how these various motives interact with each other or mediate other motives to generate gambling. It is much more complicated to find out how these motives are simultaneously mediating or interacting with one another. The excitement and avoidance motives may directly exercise a significant influence on the severity of, or high involvement in, gambling as independent variables. However, the monetary motive may exercise a more powerful influence on gambling severity in comparison to other motives (Coman et al., 1997; Neighbors et al., 2002). Many people who gamble for one reason may also gamble for another. In this case, how will the monetary motive operate in causing gambling severity? Will the amusement, avoidance and excitement motives contribute directly to gambling severity? Or will they

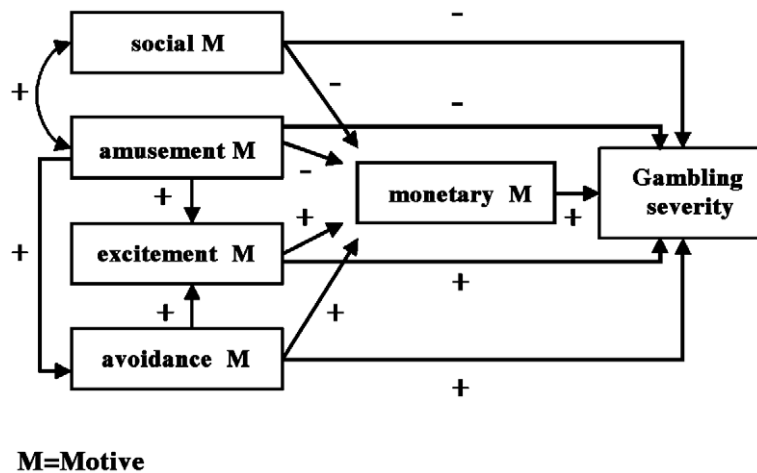


Fig. 1. Monetary motive mediation model.

indirectly influence gambling behavior through the mediation of other motives? Emotional arousal can be amplified, or unpleasant emotions can subside, depending on such incentives as monetary reinforcement or recovery of losses in gambling situations. In these circumstances, the avoidance and excitement motives can have an impact on gambling severity through the mediation of a monetary motive.

Based on this perspective, this study aimed to develop a five-factor gambling model by utilizing a self-reported questionnaire. We also sought to determine which model, parallel or monetary motive mediation, better describes the influence of gambling motives on gambling severity. The monetary motive mediation model was constructed under the hypothesis that the amusement, excitement, and avoidance motives would influence gambling severity through the mediation of the monetary motive (Fig. 1). The parallel model hypothesized that the socialization, amusement, excitement, avoidance, and monetary motives would independently influence gambling severity (Fig. 2). The purposes of this study were three-fold: (1) developing a multi-factor gambling motivation model by using college student subjects; (2) cross-validating this model on regular gamblers of varying degrees of severity; and (3) determining the effectiveness of the parallel versus monetary motive mediation model through a covariance structure equation model.

## 2. Study One

### 2.1. Overview

Study One had two purposes. The first was to construct the items appropriate for a multi-factor gambling moti-

vation model. The second was to discover the factorial structure and internal consistency of the gambling motivational scale by using exploratory factor analysis and confirmatory factor analysis.

### 2.2. Subjects

- (1) Items for the questionnaire were generated by using 34 graduate students majoring in psychology at universities in metropolitan Seoul, Korea, and 32 horse-racing gamblers recruited from Kwachon, the location of the only horse racetrack in Korea. The average age of the college participants was 28.09 years (S.D.=1.93) with a range of 25 to 32. Of the college subjects, 20 (58.8%) were male, and 14 (41.2%) were female. The horserace gamblers gave their consent to participate when they sought help for serious gambling problems. Their average age was 42.56 years (S.D.=7.52) with a range of 29

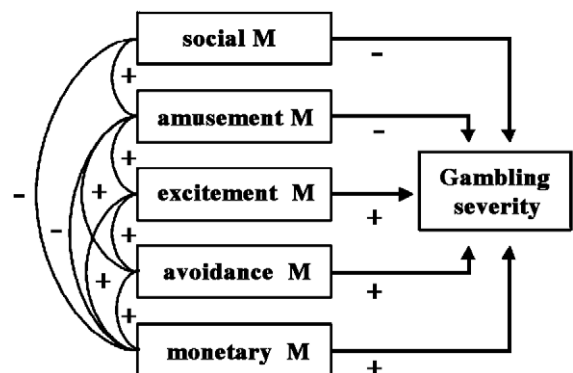


Fig. 2. Parallel model.

to 59. Twenty-six (81.3%) of them were Korean males, and six (18.7%) of them were Korean females. Seven (21.9%) were probable problem gamblers (SOGS of 3 or 4), and 25 of them (78.1%) were probable pathological gamblers (SOGS of  $\geq 5$ ).

- (2) To develop a multi-factor gambling motivational model, 240 college students were used. The participants were psychology majors over the age of 19 drawn from undergraduate colleges located in Seoul, Korea. Korean males in this group numbered 154 (64.2%), and Korean females numbered 86 (35.8%). The average age was 23.96 years (S.D.=2.26) with a range of 19 to 29.

### 2.3. Assessments and procedures

#### 2.3.1. K-South oaks gambling screen

The severity of gambling problems was assessed through the South Oaks Gambling Screen (SOGS) (Lesieur and Blume, 1987). The SOGS is a 20-item questionnaire based on the Diagnostic and Statistical Manual 3rd Edition (DSM-III) that screens for lifetime gambling problems and pathological gambling. Scores on the SOGS range from 0 to 20, and a SOGS score of 3 or 4 is taken as the cutoff to identify problem gamblers, and 5 or greater to identify pathological gamblers. The internal consistency of SOGS is 0.97, and the test–retest reliability is 0.71 (Lesieur and Blume, 1987). Recently, Choi et al. (2001) translated the SOGS into Korean, making a Korean norm and validating it: in their study, the internal consistency was reliable (0.95), with 97.2% sensitivity and 96.5% specificity. A score of 5 or higher was used as the diagnostic cutoff for pathological gambling, similar to the SOGS criteria used by Lesieur and Blume (1987). Since SOGS has been widely used to screen for the severity of gambling problems (Blaszczynski and Steel, 1998; Ladouceur et al., 1998), we decided to use the Korean edition (K-SOGS) to assess gambling severity in our study. All SOGS scores obtained in this study were obtained using the K-SOGS. All participants were also asked to report on how often they gambled and their preferred type of gambling.

#### 2.3.2. Procedure

A total of 66 subjects—the 34 graduate students and 32 horseracing gamblers—were asked to write down more than five motives of gambling in response to questions such as “Why do you think people gamble?” and “If you have ever gambled, what was the reason for gambling?” Sixty-seven items were tabulated, and related to the amusement, socialization, excitement, avoidance and monetary motives with more than 10% of the responses

from the participants were selected from the item pools. Two experienced clinical psychologists were asked to select the more relevant items by deleting overlapping and unclear items, and the resulting final 51 items were selected for further study. The 66 preliminary participants were also asked to complete the K-SOGS and to report what form(s) of gambling they participated in and how often.

The 51 selected items were administered to 240 college subjects. Five-point Likert (Likert, 1932) type scales were included for each statement, and the respondents were asked to rate the extent of their agreement with each item. Categories for the scale ranged from strongly disagree (1) to neutral (3) to strongly agree (5). Principal axes factoring with varimax was used to develop the final gambling motivation scale that met the following criteria: (a) minimum eigenvalues of 1 from scree plot tests; (b) minimum factor membership of five items; (c) elimination of items with factor loadings lower than 0.3; (d) elimination of items with communalities lower than 0.4. In addition, the internal consistency was calculated. Analyses were conducted on SPSS 11.0 for Windows.

Confirmatory factor analysis (using Analysis of Moment Structure [AMOS] 4.0 program) was conducted to determine what model would explain the data set most effectively. The five-factor gambling motive model was compared with other alternative models, which have fewer factors, by using maximum likelihood estimation. The Goodness of Fit Index was assessed by using the Bentler Bonett Normed Fit Index (NFI), the Compare Fit Index (CFI), and the Bentler Bonett Non-Normed Fit Index (NNFI). Values greater than 0.90 are regarded as acceptable (Bentler, 1995). If the value of the Root Mean Square Error of Approximation (RMSEA), which is based on population error of approximation and measures ‘discrepancy per degree of freedom’ (Joreskog and Sorebom, 1993), is 0.05 or less, then it is recognized as a close fit, and values up to 0.08 are recognized as a reasonable error of approximation.

### 2.4. Results

#### 2.4.1. K-SOGS scores and preferred forms of gambling

Among the 34 students used to develop the questionnaire, 29 (85.3%) of them were non-problem gamblers (South Oaks Gambling Screen [SOGS]=0), and five (14.7%) participants were probable social gamblers or had minimal gambling problems (SOGS of 1 or 2). The most preferred forms of gambling reported by the students were: 21 (61.8%) hwatu (Korean traditional gambling), 7 (20.6%) lottery, 3 (8.8%) poker, and 1 (2.9%) on-line gambling. In this sample, 5 (14.7%) subjects played

Table 1  
Rotated factor matrix and Cronbach's alpha of the five-factor gambling motives scale

| Items                                                    | Factor 1 | Factor 2 | Factor 3 | Factor 4 | Factor 5 | Communalities |
|----------------------------------------------------------|----------|----------|----------|----------|----------|---------------|
| Enjoy thrilling experience in risk                       | 0.837    |          |          |          |          | 0.829         |
| Have fun in risk taking                                  | 0.795    |          |          |          |          | 0.753         |
| Have fun in competing with others                        | 0.735    |          |          |          |          | 0.682         |
| Enjoy intense feelings                                   | 0.713    |          |          |          |          | 0.687         |
| Want to enjoy uncertainty                                | 0.680    |          |          |          |          | 0.555         |
| Have fun in guessing the results                         | 0.666    |          |          |          |          | 0.631         |
| Want to experience excitement and pleasure               | 0.659    |          |          |          |          | 0.613         |
| Want to feel triumph when winning                        | 0.610    |          |          |          |          | 0.452         |
| Easily absorbed in gambling                              | 0.607    |          |          |          |          | 0.590         |
| Win big money with small money                           |          | 0.802    |          |          |          | 0.716         |
| Make money easily                                        |          | 0.802    |          |          |          | 0.701         |
| Win big money immediately                                |          | 0.775    |          |          |          | 0.715         |
| Need big money                                           |          | 0.769    |          |          |          | 0.692         |
| Heard that they won jackpot                              |          | 0.734    |          |          |          | 0.594         |
| Have a financial difficulty and no money                 |          | 0.688    |          |          |          | 0.679         |
| Can't change my life without gambling                    |          | 0.582    |          |          |          | 0.464         |
| May win big money                                        |          | 0.557    |          |          |          | 0.486         |
| Feel pain/troubled                                       |          |          | 0.773    |          |          | 0.703         |
| Feel lonely/escape from loneliness                       |          |          | 0.738    |          |          | 0.652         |
| Feel depressed/sad                                       |          |          | 0.733    |          |          | 0.638         |
| Feel angry/upset                                         |          |          | 0.730    |          |          | 0.689         |
| Feel tense/anxious                                       |          |          | 0.678    |          |          | 0.547         |
| Feel pressured/things don't go well                      |          |          | 0.635    |          |          | 0.595         |
| Forget about stressful reality                           |          |          | 0.616    |          |          | 0.599         |
| Socialize with others                                    |          |          |          | 0.755    |          | 0.629         |
| Make the atmosphere comfortable for meeting people       |          |          |          | 0.736    |          | 0.599         |
| Makes it easy to meet new people                         |          |          |          | 0.688    |          | 0.487         |
| Join with gathering in spite of no intention of gambling |          |          |          | 0.647    |          | 0.466         |
| Friends insisted gambling                                |          |          |          | 0.645    |          | 0.506         |
| Get along with others favorably                          |          |          |          | 0.619    |          | 0.533         |
| Change moods                                             |          |          |          |          | 0.686    | 0.638         |
| Escape from burdensome routines                          |          |          |          |          | 0.542    | 0.572         |
| Energize life                                            |          |          |          |          | 0.472    | 0.521         |
| Enjoy leisure time and activity                          |          |          |          |          | 0.432    | 0.378         |
| Relieve stress                                           |          |          |          |          | 0.377    | 0.452         |
| Eigenvalue                                               | 12.63    | 4.71     | 2.47     | 2.06     | 1.20     |               |
| Cumulative variance explained (%)                        | 36.08    | 49.54    | 56.59    | 62.48    | 65.91    |               |
| Cronbach's $\alpha$                                      | 0.94     | 0.85     | 0.91     | 0.92     | 0.80     |               |

gambling games more than 5 times a month, 2 (5.9%) subjects 3 to 4 times a month, 3 (8.8%) subjects 1 to 2 times a month, and 22 (64.7%) subjects less than once a month. Among the 34 horserace gamblers, 7 (21.9%) were probable problem gamblers (SOGS of 3 or 4) and 25 of them (78.1%) were probable pathological gamblers (SOGS of  $\geq 5$ ).

Among the 240 students used to produce the model, the number of students in each SOGS category was: 199 (82.9%) non-problem gamblers (SOGS=0), 35 (14.6%) social or minimal problem gamblers (SOGS of 1 or 2), 3 (1.2%) probable problem gamblers (SOGS of 3 or 4), and 3 (1.2%) probable pathological gamblers (SOGS of  $\geq 5$ ). Among them, 157 were current gamblers: hwatu 93

(38.8%), poker 31 (12.9%), lottery 23 (9.6%), stock market 2 (.8%), and horseracing 1 (.4%); 83 (34.6%) were past gamblers and 6 (2.5%) did not complete the SOGS. Among current gamblers, 124 (51.7%) played gambling games less than once a month, 22 (9.2 %) two to four times a month, and 10 (4.2%) more than 5 times a month.

#### 2.4.2. Exploratory factor analysis

Among the 51 items, items with communalities of 0.4 or less and items with less than 0.10 difference of factor loading between two factors were eliminated. After eliminating items according to selection criteria, 35 items were selected (Table 1). The Kaiser–Meyer–Olkin measure of sampling adequacy (0.93) and Barrett's Test of Sphericity

( $\chi^2=5435.87$ ,  $P<0.001$ ) each indicated that the correlation matrix was appropriate for such an analysis.

Five factors with eigenvalues greater than 1.0 were extracted from the matrix (Table 1). Examination of the scree plot suggested that five factors were most appropriate. All 35 items had factor loadings of greater than 0.35 and communalities of greater than 0.40 except one item (i.e., enjoy leisure time and activity). None of the items had substantial cross-loading on other factors. The five factors accounted for 65.91% of the variance in scores: factor one accounted for 36.08% variance with eigenvalue=12.63; factor two, 13.45% with eigenvalue=4.71; factor three, 7.06% with eigenvalue=2.47; factor four, 5.89% with eigenvalue=2.06; and factor five, 3.43% with eigenvalue=1.20.

Factor one relates to the thrill, tension, and excitement experienced from gambling and was named as the excitement motive. Factor two consists of items related to social gatherings, interactions, and enjoying the social atmosphere through gambling. It was labeled the socialization motive. Factor three was named the avoidance motive because all of its items represent the gamblers' motivation to reduce negative feelings such as low self-esteem and unpleasant moods. Factor four represents the gamblers' motivation for hitting a jackpot to make big, easy money and was named the monetary motive. Factor five relates to gaining fun, enjoyment, and pleasure from gambling and was named the amusement motive.

#### 2.4.3. Confirmatory factor analysis

Confirmatory factor analysis was conducted to determine which model better explained the data set. The five-factor gambling model was compared to the five alternative motivational models by using maximum likelihood estimation. The one-factor model attempted to explain gambling motives with a common factor. The two-factor model was set up as one-factor under the assumption that the correlation between the socialization and amusement motives is very high. Both motives will not positively affect gambling severity or prevent gambling problems. The rest of the three motives were combined as one factor. The three-factor model collapsed the excitement and avoidance motives as one factor because both factors are highly correlated with each other and are anticipated to provide positive and negative emotional reinforcements. It was assumed that the social and amusement motives were the second factor, and the monetary motive was the third motive in the three-factor model. The four-factor model was formulated as A and B models. Model A combined the socialization and amusement factors as one factor because both of them are anticipated to reduce the severity of gambling problems. The rest of the four factors

were regarded as independent factors. Model B treated amusement and excitement as one factor because both factors are anticipated as emotional and pursuant of positive reinforcement. If the Fit Index of the five-factor model is not higher than that of other alternative models, then the five-factor model will be rejected, and a model with fewer factors will be selected based on parsimony.

The Fit Indices of the alternative motive models and five-factor models were examined, and the Fit Index value of the five-factor model was the highest. The one-, two- and three-factor models' Fitness Indices were less than the accepted value of 0.90. The RMSEA values were also beyond the acceptable range (for the one-factor model, NFI=0.79, CFI=0.82, NNFI=0.80, RMSEA=0.14; for the two-factor model, NFI=0.82, CFI=0.85, NNFI=0.83, RMSEA=0.12; for the three-factor model, NFI=0.86, CFI=0.90, NNFI=0.89, RMSEA=0.10). For the A and B types of the four-factor model, the Relative Fit Indices were acceptable ( $>0.90$ ), and RMSEA values were also acceptable (four-factor model A: NFI=0.90, CFI=0.94, NNFI=0.93, RMSEA=0.08; four-factor model B: NFI=0.90, CFI=0.94, NNFI=0.93, RMSEA=0.08). However, the Relative Fit Index was highest for the five-factor motivational model (NFI=0.92, CFI=0.95). The NNFI and RMSEA indices, which consider the principle of parsimony, showed especially that the Fit Index of the five-factor model was the highest (NNFI=0.95, RMSEA=0.07). Therefore, the five-factor model was considered the most appropriate model when considering the parsimony principle.

#### 2.4.4. Reliability and inter-correlation factors for the five-factor model

Cronbach's alpha for the five-factor gambling motivational scale was high (alpha=0.94). Factor one consisted of nine items; factor two consisted of eight items; factor three consisted of seven items; factor four consisted of six items; and factor five consisted of five items.

The Pearson product moment correlation coefficients between all of the factors were significant, except for the correlation between the socialization and monetary motives ( $r=0.06$ , ns). All of the correlations between the various motives were significant: between the excitement and monetary motives,  $r=0.53$  ( $P<0.01$ ); between the excitement and avoidance motives,  $r=0.57$  ( $P<0.01$ ), between the excitement and amusement motives,  $r=0.59$  ( $P<0.01$ ); between the excitement and socialization motives,  $r=0.37$  ( $P<0.01$ ); between the avoidance and monetary motives,  $r=0.54$  ( $P<0.01$ ); between the avoidance and amusement motives,  $r=0.51$  ( $P<0.01$ ); between the socialization and amusement motives,  $r=0.54$  ( $P<0.01$ ); between the socialization and avoidance

motives,  $r=0.18$  ( $P<0.01$ ); and between the amusement and monetary motives,  $r=0.25$  ( $P<0.01$ ).

### 3. Study Two

#### 3.1. Overview

Since the five-factor gambling motive model extracted in Study One was confirmed by using college students, it may not be applicable to regular gamblers. To apply this model to regular gamblers, Study Two was conducted with the following purposes in mind: cross-verifying the five factor gambling motive model with regular gamblers who are currently participating in gambling, and comparing the effectiveness of a monetary motive mediation model with a parallel model to describe how gambling motives affect gambling severity.

#### 3.2. Subjects

The subjects were 240 volunteers who were actively participating in gambling activities; many in this group were suffering from legal or psychological problems due to their gambling activities. They were recruited with their permission from the Kwachon Horse Racecourse in metropolitan Seoul and a casino in Kangwon Province where native customers are allowed. (Typically, native customers are not allowed to enter Korean casinos.) They were also recruited from individuals who were admitted to a psychiatric unit for gambling treatment or who were under probation for illegal gambling activities and ordered to complete remedial education. All the subjects were paid for their participation and were given information about the purpose of our study. The average age of the subjects was 38.94 years ( $S.D.=8.20$ ), with a range of 22 to 70 years. The number of Korean males was 203 (84.19%), and of Korean females was 37 (15.81%).

#### 3.3. Procedure

To test the suitability of the five-factor model, principal axes factoring with varimax was conducted, and internal consistency was calculated. Confirmatory factor analysis was performed to evaluate the fitness of our models like we did in Study One.

Path analysis, by using covariance equation structure modeling, was conducted to assess the fitness and magnitude of the monetary motive mediation model's influence compared to that of the parallel model's in order to determine which model better explained the effect of gambling motives on gambling severity. Maximum likelihood estimation was used, and the Goodness of Fit Index

was assessed by using GFI, NFI, CFI, NNFI, and RMSEA. Gambling severity was assessed with K-SOGS.

#### 3.4. Results

##### 3.4.1. K-SOGS scores and preferred forms of gambling

Overall, 19 (7.9%) of the gamblers were non-problem gamblers (SOGS=0), 23 (9.6%) were social gamblers (SOGS of 1 or 2), 25 (10.4%) were probable problem gamblers (SOGS of 3 or 4), and 173 (72.1%) probable pathological gamblers (SOGS of  $\geq 5$ ). When the main gambling was horseracing, there were: 12 (12.0%) non-problem gamblers (SOGS=0), 12 (12.0%) social gamblers (SOGS of 1 or 2), 12 (12.0 %) probable problem gamblers (SOGS of 3 or 4), and 64 (64.0%) probable pathological gamblers (SOGS of  $\geq 5$ ). When casinos were the main form of gambling, there were: 1 (1.3%) social gambler, 4 (5.2 %) probable problem gamblers, and 71 (93.4%) probable pathological gamblers. When poker was the main form of gambling, there were: 1 (1.7%) non-problem gambler, 4 (10.8%) social gamblers, 5 (13.5 %) probable problem gamblers, and 27 (73.0%) probable pathological gamblers. When hwatu was the main form of gambling, there were: 3 (25.0%) non-problem gamblers, 4 (33.3%) social gamblers, and 5 (41.7%) probable pathological gamblers. When lottery was the main form of gambling, there were: 1 (10.0%) non-problem gambler, 2 (20.0%) social gamblers, 3 (30.0%) probable problem gamblers, and 4 (40.0%) probable pathological gamblers. When cycle racing, slot machines and stocks were taken together, there were: 2 (40.0%) non-problem gamblers, 1 (20.0 %) probable problem gambler, and 2 (40.0%) probable pathological gamblers.

In most cases, the subjects in this study were multiple gamblers, but the breakdown of the subjects' primary gambling activities was as follows: 100 of the subjects (41.7%) were engaging in horserace gambling; 76 (31.7%) were spending time in the casino; 37 (15.4%) were playing poker; 12 (5.0%) were involved in hwatu, Korean traditional gambling; 10 (4.2%) were playing the lottery; one (0.4%) betted on cycling races; one (0.4%) play slot machine; and three (1.3%) were playing the stock market heavily.

##### 3.4.2. Confirmatory factor analysis and internal consistency

Confirmatory factor analysis was conducted to determine which model explained the data set more effectively. By using maximum likelihood estimation, the five alternative motivation models were compared with the five-factor gambling motivation model. The results showed that Fit Index value for the five-factor model

was the highest. All fitness indices were less than the accepted value of 0.90 in the one-, two-, and three-factor models. RMSEA values were beyond the acceptable values for the one-factor model (NFI=0.82, CFI=0.85, NNFI=0.83, RMSEA=0.13), the two-factor model (NFI=0.85, CFI=0.88, NNFI=0.86, RMSEA=0.12), and the three-factor model (NFI=0.88, CFI=0.91, NNFI=0.90, RMSEA=0.10). With regards to the A and B types of the four-factor model, the Relative Fitness Indices were above 0.90 and RMSEA was in the acceptable range (for type A: NFI=0.91, CFI=0.94, NNFI=0.94, RMSEA=0.08; for type B: NFI=0.91, CFI=0.95, NNFI=0.94, RMSEA=0.08). However, the Relative Fit Index value was the highest in the five-factor model (NFI=0.92, CFI=0.96). When the principle of parsimony was also considered, the fitness of the five-factor model was the highest (NNFI=0.95, RMSEA=0.07).

Cronbach's alpha for the five-factor gambling motivational scale was high ( $\alpha=0.92$ ). Moderate to high reliability was also found for each of the five subscales: excitement motive ( $\alpha=0.89$ ); monetary motive ( $\alpha=0.87$ ), avoidance motive ( $\alpha=0.90$ ), socialization motive ( $\alpha=0.83$ ), and amusement motive ( $\alpha=0.78$ ).

### 3.4.3. A comparison of the parallel and monetary motive mediation models

The fitness of the parallel model and monetary motive mediation model was analyzed by using covariance structure equation modeling. In this model, the correlation between the socialization motive and amusement motive was statistically significant ( $r=0.42$ ,  $P<0.01$ ), but the socialization motive did not exert meaningful influences on the monetary motive ( $\beta=-0.078$ , ns) or gambling severity ( $\beta=-0.037$ , ns). Therefore, a revised model was suggested, shown in Fig. 3, after the effect of the socialization motive was corrected.

For the parallel model, the GFI and CFI were in an acceptable range (GFI=0.95, CFI=0.90), but the NFI did not meet the standard (NFI=0.89) and the NNFI and RMSEA values fell very short (NNFI=0.61, RMSEA=0.21). In contrast, the values of GFI, NFI, and CFI for the monetary motive mediated model were above the 0.90 level, and all of them were within an acceptable range (GFI=0.98, NFI=0.97, CFI=0.98). The NNFI related to parsimony was satisfactory (NNFI=0.91). Although RMSEA fell short of the 0.80 level (RMSEA=0.10), the monetary motive mediated model seems to be a relatively more valid and superior model compared to the parallel model.

In the modified monetary motive mediated model, we examined a hypothetical relationship among the variables. The amusement motive exercised a positive influence over the avoidance motive ( $\beta=0.41$ ,  $P<0.001$ ), and, together with the avoidance motive, also influenced the excitement motive positively ( $\beta=0.44$ ,  $P<0.001$ ;  $\beta=0.27$ ,  $P<0.001$ ). The 16.5% variance of the avoidance motive was explained by the amusement motive. In regard to the excitement motive, 35.7% of the excitement variance was explained by the amusement motive and the avoidance motive. In terms of the monetary motive, 34.6% of the variance was explained by the avoidance motive, the excitement motive, and the amusement motive. Among them, the influence of the excitement motive was the strongest ( $\beta=0.45$ ,  $P<0.001$ ), followed by the avoidance motive ( $\beta=0.39$ ,  $P<0.001$ ). On the other hand, the amusement motive exerted direct, yet negative, influence on the monetary motive ( $\beta=-0.38$ ,  $P<0.001$ ). That is, the monetary motive became stronger as the avoidance and excitement motives became stronger. However, as the amusement motive became stronger, the monetary motive became less influential. The excitement, avoidance, and amusement motives were able to explain 35.7% of the monetary motive's variance. Only the

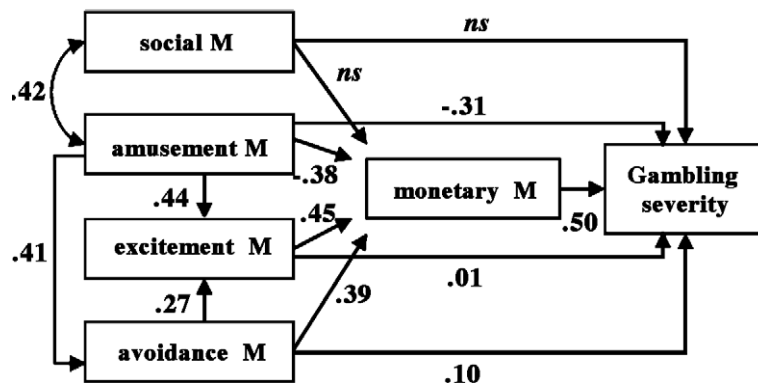


Fig. 3. Modified monetary motive mediation model.

Table 2

Magnitude of indirect and direct effects of gambling motives' influence on gambling severity

|                    | Amusement/<br>excitement | Amusement/<br>avoidance | Amusement/<br>monetary | Amusement/<br>G severity | Avoidance/<br>excitement | Avoidance/<br>monetary | Avoidance/<br>G severity | Excitement/<br>monetary | Excitement/<br>G severity | Monetary/<br>G severity |
|--------------------|--------------------------|-------------------------|------------------------|--------------------------|--------------------------|------------------------|--------------------------|-------------------------|---------------------------|-------------------------|
| Total<br>effect    | 0.55                     | 0.41                    | 0.02                   | −0.25                    | 0.27                     | 0.51                   | 0.36                     | 0.45                    | 0.24                      | 0.50                    |
| Causal<br>effect   |                          |                         |                        |                          |                          |                        |                          |                         |                           |                         |
| Direct<br>effect   | 0.44                     | 0.41                    | −0.38                  | −0.31                    | 0.27                     | 0.39                   | 0.10                     | 0.45                    | 0.01                      | 0.50                    |
| Indirect<br>effect | 0.11                     | 0.00                    | 0.40                   | 0.06                     | 0.00                     | 0.12                   | 0.26                     | 0.00                    | 0.23                      | 0.00                    |

G=gambling.

monetary and amusement motives exerted significant influence over the severity of gambling. The monetary motive showed a positive contribution to the severity of gambling ( $\beta=0.50$ ,  $P<0.001$ ), while the amusement motive showed a negative contribution to the severity of gambling ( $\beta=-0.31$ ,  $P<0.001$ ). This means that the severity of gambling rose with the strength of the monetary motive, but when the amusement motive became stronger, gambling severity decreased. In contrast, the avoidance and excitement motives did not exert direct influences on gambling severity ( $\beta=0.10$ , ns;  $\beta=0.01$ , ns).

The modified monetary motive mediated model is outlined in Fig. 3. The process by which gambling motives influence gambling severity in the direct and indirect effects of the modified model is detailed in Table 2. In this modified model, only the monetary motive exercised a direct positive influence on gambling severity, while the avoidance and excitement motives showed a more indirect influence (0.26, 0.23), rather than a direct influence (0.10, 0.01), on gambling severity. In contrast, the excitement and avoidance motives had a positive influence on the monetary motive (0.39, 0.45) and thus exerted an indirect influence on gambling severity through the monetary motive. The amusement motive showed a negative influence (−0.25) on gambling severity. Further analysis of the amusement motive, in terms of direct and indirect effects, showed a direct negative (−0.31) influence on gambling severity. However, it demonstrated an indirect positive effect of 0.06 on gambling severity through the excitement (0.44) and avoidance (0.41) motives.

#### 4. Discussion

This study attempted to develop a comprehensive gambling motive model and compare the monetary motive mediated model and parallel model for explaining how gambling motives affect the severity of gambling. First, the five-factor gambling motive model, which consisted of the socialization, amusement,

excitement, avoidance, and monetary motives, was developed through both exploratory and confirmatory factor analysis using college subjects. The internal consistency of the total scale was 0.94. The five-factor motive model demonstrated the highest fitness compared to other models. Second, the same factorial structure was also confirmed through both exploratory and confirmatory factor analysis among frequent gamblers. Third, the results showed that the monetary motive mediation model was more effective than the parallel model in explaining the influence of gambling motives on gambling severity. The social motive was eliminated because it had no effect on the monetary motive and gambling severity. In the modified mediation model, the monetary motive showed positive direct influences on gambling severity. The avoidance motive and excitement motive showed no direct influences on gambling severity. Instead, they exerted an indirect influence on gambling severity through the monetary motive. Furthermore, the avoidance motive also influenced gambling severity indirectly through the excitement motive.

These findings suggest that gamblers have a strong desire to win a big jackpot in real gambling situations and that unrealistic beliefs or their desire for big money can trigger a vicious cycle of gambling involvement. Although the excitement and avoidance motives may influence gambling problems, they facilitate gambling severity through mediation of the monetary motive. The gamblers may not be aware of how the avoidance and excitement motives are operating in recurrent gambling situations because they are so preoccupied with the immediate and visible gain of winning money. The pathological gambler has a dream of gaining immediate wealth and believes he or she can win a lot of money easily (Custer, 1984).

Our results may look inconsistent with existing studies, which show that the purposes of gambling include stimulation, relaxation and escape from emotional tension (Anderson and Brown, 1984; Leary and

Dickerson, 1985; Brown, 1986; Jacobs, 1987; Blaszczynski and Steel, 1998). However, this result does not mean that the avoidance and excitement motives do not impact gambling problems or their severity. Individuals gamble in order to maintain an optimal state of arousal (Brown, 1986). People also take risks to bet on uncertainties or induce arousal in order to compensate or convert negative emotions to positive moods (Raghunathan and Pham, 1999), but various motives are operating simultaneously in real gambling situations. We speculate that the influence of the avoidance and excitement motives work, or are amplified, through the reinforcement of monetary gain or recovery of financial loss. Gambling is different from alcohol in terms of external reinforcement. In order to enable emotional arousal or avoidance in gambling, there must be big, but uncertain, monetary reinforcements, and individuals have to place a bet on this uncertain reward. According to recent studies (Breiter et al., 2001; Kuhn and Knutson, 2005; Knutson et al., 2005), positively aroused feelings associated with anticipation of monetary gain (e.g., “excitement”) may promote risk taking, and the nucleus accumbens is activated during risky choices that may bring a large monetary reward.

The socialization and amusement motives contributed to healthy and sociable gambling, or, at least, they did not directly influence gambling severity. The amusement motive functions positively or negatively, depending on the direct or indirect effect. In the beginning, the amusement motive appears to counteract against monetary motivations and gambling severity. However, the amusement motive may indirectly facilitate the avoidance and excitement motives. They may, therefore, contribute to gambling severity on only a mild level. If the gambler has a strong amusement motivation, (s)he may be likely to pursue an intense and immediate state of excitement through gambling. This helps to counteract negative feelings such as depression and low self-esteem.

Most gamblers have won big money during an early phase of gambling (Ladouceur and Mayrand, 1984) and they believe that winning again is the only way to relieve themselves of the stress of their losses (Custer, 1984; Hercovitch, 1999). Pathologic gamblers have a tendency to choose easy but risky ways to earn money, to regard the kind of achievement, reward, and delayed gratification gained from hard work as inferior means (Lesieur, 1992b). They also tend to deny the fact that their difficulties in life are associated with gambling, and to have a continued belief that they will win big money (Gabour and Ladouceur, 1988; Moore and Ohtsuka, 1999; Toneatto, 1999). Therefore, therapists

need to understand the gambler's wrong beliefs and eventually help them to abandon their pathological expectations and achieve rational alternatives.

The fact that our subjects were all a single ethnic group, native Korean, poses both an advantage and a limitation to our study. On the one hand, working with a single ethnic group may simplify the analysis of the study and furthermore, may shed some light on cultural influences on gambling behavior. There may be interesting cultural effects on gambling motives. For example, Oriental culture, including Korean, considers interdependence between individuals more important than individualism, while Western culture emphasizes individualism rather than collectivism (Hofstede, 2004; Triandis, 1995). Games and gambling appear to play a role in the enhancement of intimacy and cohesion between group members in a collectivist society (i.e., “making the atmosphere more comfortable for meeting people”). On the other hand, the single culture of our study may limit its applicability to other cultures. Some details in the content or expression of each factor could be different according to cultural differences. For this reason, the generalizations of our study need further replication in other cultures. However, given that Cooper's (1992) three-factor structure for drinking motivation was replicated in Koreans (Sin, 1999) and that several studies conducted in western societies also reported that experiencing excitement and winning money were the most important gambling motives (Coman et al., 1997; Mississippi State University Gambling Group, 1995; Neighbors et al., 2002), it seems that cultural differences may not change the main factor structure that we found.

Another limitation of our study was that non-problem gamblers in Study Two were included. The reason for including non-problem gamblers was so that we could identify gambling motives and the severity of gambling among individuals in the entire continuum from non-problem to pathological gambling. Subsequent studies, however, will be necessary to assess and attain more specific and concrete data exclusively from problematic or pathological gamblers. Among the gamblers, the frequency, period, and amount of gambling was not evaluated either. It is possible that there could be more or less than five gambling motives. Chantal et al. (1994, 1995) developed a gambling motivation scale based on self-determination theory and the tripartite taxonomy of intrinsic motivation. They differentiated between an intrinsic motivation to experience stimulation, an intrinsic motivation to know, and an intrinsic motivation to accomplish. They further classified three types of internal and external motivations as external regulation,

introjected regulation, and identified regulation. Since our study consists of empirical research, intrinsic motivations were not included in our study. In subsequent studies, how intrinsic motivations lead to gambling severity may need to be examined.

In conclusion, our results suggest that the monetary motive may mediate other motives for gambling behavior. Therefore, professionals should not underestimate the determinant role of monetary reward in the pathogenesis of pathological gambling in order to develop a realistic treatment and prevention program for pathological gambling.

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